Function-Based and Reverse RSA Encryption Systems

Amine Belachhab - HS Independent Researcher

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Abstract

This paper presents two novel encryption systems designed to address potential vulnerabilities in classical RSA arising from advances in quantum computing and mathematical insights from the Riemann Hypothesis. The proposed systems—Function-based RSA (fRSA) and Reverse RSA (rRSA)—introduce function-based transformations and precision-dependent security mechanisms that move beyond traditional prime factorization hardness. Each system offers three implementation variants (Standard, Super, and Hybrid), providing flexible security levels adaptable to different applications, from casual messaging to military communications.

1 Introduction

Classical RSA security relies on the computational difficulty of factoring large composite numbers into their prime factors. However, potential breakthroughs in quantum computing (Shor's algorithm) and deeper mathematical understanding of prime distribution through the Riemann Hypothesis pose existential threats to this foundation.

This paper proposes a fundamental reconceptualization of public-key cryptography that maintains the mathematical elegance of prime-based systems while introducing novel hardness assumptions. Rather than relying solely on integer factorization, these systems incorporate function-based transformations and precision-dependent security mechanisms that create multiple layers of computational difficulty for potential attackers.

2 Part 1: Function-Based RSA (fRSA)

2.1 System Overview

fRSA operates on the principle that instead of publishing the direct product of two primes, the public key consists of the product of their images under a secret transformation function. This approach preserves the prime-based foundation while obscuring the underlying mathematical structure through functional composition.

2.2 Key Generation

Private Components:

- Two large primes: a, b
- Secret transformation function: $f(x)$ with real coefficients
- Precision parameter: d (number of decimal places for security)

Public Components:

- Modulus: $N = a \times b$
- Transformed key: $K_{pub} = f(a) \times f(b)$ (published to d decimal places)

Key Generation Process:

1. Select cryptographically strong primes a, b
2. Generate transformation function $f(x)$ using one of three methods (see variants)
3. Compute $K_{pub} = f(a) \times f(b)$
4. Truncate K_{pub} to d decimal places based on required security level
5. Publish (K_{pub}, N) as public key

2.3 Encryption and Decryption

Encryption: $c = m^{K_{pub}} \bmod N$
Decryption: $m = c^{(1/K_{priv})} \bmod N$
Where K_{priv} is the full-precision value of $f(a) \times f(b)$ known only to the key owner.

2.4 Security Model

The security of fRSA rests on three computational challenges:

1. **Function Recovery:** Determining $f(x)$ from public information
2. **Prime Factorization:** Finding a, b from $N = a \times b$
3. **Precision Attack:** Determining K_{priv} to sufficient decimal precision

Attack complexity: $O(|F| \times 2^{|P|} \times 10^d)$ where $|F|$ is the function space size, $|P|$ is the prime bit-length, and d is the precision parameter.

2.5 fRSA Variants

2.5.1 Standard fRSA

- Function f is exchanged once through secure channel
- Remains static until manual update
- Suitable for long-term key pairs

2.5.2 Super fRSA

- Function f generated deterministically from shared seed
- Periodic regeneration (per session/monthly)
- No transmission of f required after initial seed exchange

2.5.3 Hybrid fRSA

- Combines seed-based generation with secure exchange
- Function generation rules updated periodically
- Balances security with practical key management

3 Part 2: Reverse RSA (rRSA)

3.1 System Overview

rRSA inverts the secrecy model of fRSA: instead of hiding the primes and revealing the transformed product, rRSA publishes the primes while keeping both the transformation function and its computed result secret. This creates a dual-secret system where attackers must overcome multiple independent computational challenges.

3.2 Key Generation

Public Components:

- Two large primes: a, b

Private Components:

- Secret transformation function: $f(x)$
- Secret key: $K_{sec} = f(a) \times f(b)$ (computed to full precision)
- Working key: $K_{work} = K_{sec}$ truncated to d decimal places

Key Generation Process:

1. Select and publish cryptographically strong primes a, b
2. Generate secret transformation function $f(x)$
3. Compute $K_{sec} = f(a) \times f(b)$ with maximum precision
4. Derive K_{work} by truncating K_{sec} to security-appropriate precision d

3.3 Encryption and Decryption

Encryption: $c = m^{K_{work}} \bmod (a \times b)$

Decryption: $m = c^{(1/K_{work})} \bmod (a \times b)$

Both parties must compute K_{work} to identical precision for successful communication.

3.4 Security Model

rRSA security depends on two independent computational problems:

1. **Function Discovery:** Determining $f(x)$ from known inputs a, b but unknown output
2. **Precision Guessing:** Computing $K_{sec} = f(a) \times f(b)$ to exact working precision

Attack complexity: $O(|F| \times 10^d)$ where $|F|$ is the function space and d is the precision requirement.

Key Insight: Even if an attacker determines $f(x)$, they must still compute K_{work} to the exact precision used by legitimate parties—an exponentially difficult precision-guessing problem.

3.5 rRSA Variants

Implementation variants mirror fRSA structure but focus on function synchronization rather than prime secrecy.

4 Part 3: Precision-Based Security Architecture

4.1 Adaptive Security Levels

Both systems implement tunable security through precision parameters:

- **Level 1 ($d = 100$):** Consumer applications (messaging, email)
- **Level 2 ($d = 500$):** Commercial/financial transactions
- **Level 3 ($d = 1000+$):** Military/government communications

4.2 Precision Attack Resistance

For precision parameter d, attackers must determine the secret key K to accuracy 10^{-d}. This creates:

- **Brute force complexity:** $O(10^d)$ trials
- **Quantum resistance:** Even Grover's algorithm provides only $O(10^{d/2})$ improvement
- **Scalable security:** Increasing d exponentially increases attack difficulty

4.3 Implementation Considerations

Synchronization Requirements:

- Function f must be identically computed by both parties
- Key K must be truncated to identical precision

- Error tolerance within precision bounds prevents communication failure

Computational Efficiency:

- Function evaluation: $O(\log n)$ for polynomial functions
- Precision arithmetic: $O(d)$ additional computational cost
- Acceptable overhead for exponential security gain

5 Part 4: Comparative Security Analysis

5.1 Classical RSA vs Proposed Systems

System	Classical Security	Hardness Assumption	Quantum Vulnerability	Performance	Scalable Security
RSA	Baseline	Integer factorization	High (Shor's algorithm)	Baseline	Fixed
fRSA	Maximum	Function recovery + factorization + precision	Medium	Slower	Yes
rRSA	High	Function discovery + precision	Low	Faster	Yes

Classical Security Ranking: fRSA > rRSA > RSA

- **fRSA**: Three independent secrets (primes + function + precision) provide maximum classical security
- **rRSA**: Two independent secrets (function + precision) offer high but reduced classical security
- **RSA**: Single hardness assumption (factorization) provides baseline security

Performance Trade-off: rRSA > fRSA > RSA

- **rRSA**: Public primes enable faster key operations and validation
- **fRSA**: Hidden primes require additional computational overhead
- **RSA**: Standard performance baseline

5.2 Post-Quantum Assessment

Quantum Attack Vectors:

- **Shor's Algorithm:** Not directly applicable (no pure integer factorization)
- **Grover's Algorithm:** Provides quadratic speedup for function/precision search
- **Period Finding:** Potentially applicable to certain function classes

Quantum Resistance: Both systems demonstrate improved quantum resistance compared to classical RSA, with security primarily dependent on function space size and precision requirements rather than integer factorization.

6 Part 5: Function Construction Guidelines

6.1 Recommended Function Classes

Polynomial Functions: $f(x) = \sum_{i=0}^{n} c_i \times x^i$

- Advantages: Efficient computation, well-understood mathematics
- Security: Depends on coefficient secrecy and degree

Transcendental Functions: $f(x) = a \cdot \log_b(c) \cdot x^2 + \sin(d) \cdot x + e$

- Advantages: Infinite coefficient space, irrational outputs
- Security: Enhanced by transcendental coefficient relationships

Composite Functions: $f(x) = g(h(x))$ where g, h are simpler functions

- Advantages: Exponential function space growth
- Security: Multiple layers of functional transformation

6.2 Security Requirements for Functions

1. **Non-invertibility:** Given $f(a) \times f(b)$, determining f should be computationally infeasible
2. **Coefficient Independence:** Function parameters should be cryptographically independent
3. **Precision Stability:** Small changes in input should not cause catastrophic precision loss
4. **Computational Efficiency:** Function evaluation should remain practically feasible

7 Conclusion

The proposed fRSA and rRSA systems represent a paradigm shift in public-key cryptography, moving beyond reliance on single hardness assumptions toward multi-layered security architectures. By incorporating function-based transformations and precision-dependent security mechanisms, these systems offer:

1. **Enhanced quantum resistance** through novel computational challenges
2. **Scalable security levels** adaptable to application requirements
3. **Mathematical flexibility** in function selection and precision tuning
4. **Practical implementability** through systematic variant structures

While these systems require further cryptanalytic evaluation and implementation optimization, they provide a promising foundation for post-quantum cryptographic security that maintains the elegance and efficiency of prime-based mathematics while addressing the fundamental vulnerabilities of classical RSA.

The precision-based security model, in particular, offers a novel approach to creating exponentially difficult computational challenges that remain intractable even under quantum attack scenarios. This work opens new research directions in adaptive cryptographic security and function-based transformation systems.

Future Work: Detailed implementation specifications, formal security proofs, and comparative performance analysis with established post-quantum cryptographic systems remain essential next steps for practical deployment.